ANDRÉ ROGÉRIO DA SILVA FILHO

Software Development Consultant | Full-Stack • Autonomous AI Agents • Cloud Architecture

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SUMMARY

Software Development Consultant specializing in full-stack architecture, autonomous AI agents, and cloud infrastructure, with a strong engineering foundation established in corporate IT operations and systems. Proven track record designing and delivering mission-critical production platforms using TypeScript, Python, React, Next.js, Node.js, and PostgreSQL. At Grupo CBI, progressed from Full-Stack Developer to Software Development Consultant, leading enterprise software engineering, autonomous AI agent architecture (Google Gemini, MCP servers, advanced RAG), cloud infrastructure on GCP (Cloud Run, Cloud SQL, Secret Manager), and high-availability relational databases safeguarding sensitive citizen data across metropolitan concessions under strict LGPD compliance. In technological research (UniFacens/MCTI/Softex), ranked 8th nationwide, engineering industrial digital twins, real-time WebSocket telemetry, and cognitive memory architectures with 3D spatial interfaces.

TECHNICAL SKILLS

Languages & Runtimes: Python, TypeScript, JavaScript, SQL, Node.js

Software Architecture & Web: React, Next.js, Fastify, Express, WebSockets, REST APIs

Applied AI & Agents: Autonomous Agents (MCP, LangGraph), LLMs (Google Gemini API), Vision Models (vLLM on AMD ROCm), Advanced RAG, NetworkX

Cloud, Data & Security: Google Cloud (Cloud Run, Cloud SQL), Docker, PostgreSQL (Row-Level Security), Redis, CI/CD, AppSec (CAP)

PROJECTS

Cribum (AuditRx) | Multimodal Medical Audit & Fraud Graph

GitHub

AMD Developer Hackathon & Campus Mobile • Python, FastAPI, React 19, PyTorch, AMD ROCm, vLLM, Qwen2-VL, NetworkX, TypeScript

- Architected a privacy-first, 4-stage multi-agent pipeline (Reader, Pharmacist, Investigator, Reporter) for prescription auditing, LASA drug safety (ISMP/ANVISA), and healthcare fraud detection, submitted to AMD Developer Hackathon and Campus Mobile.
- Integrated local Vision-Language Models (Qwen2-VL via vLLM on AMD ROCm) with uncertainty-aware multi-hypothesis extraction, clinical risk scoring (CCRI), and an RFC 6962 Merkle Tree custody chain for strict LGPD compliance.
- Engineered a NetworkX multigraph anomaly engine over 200+ synthetic claims, validated against a custom Ground Truth benchmark (LASA ambiguity, CID/CRM mismatches, duplicate claims) and a 36-test suite (22 PyTest and 14 Playwright E2E scenarios).

FundiVR | Industrial Digital Twin & 700°C+ Metallurgy VR Simulator

GitHub

Digital Twin & 5G Residency (MCTI / Softex / UniFacens) • TypeScript, React, Zustand, Fastify, PostgreSQL, Prisma, WebSockets, Zod, pnpm Monorepo

- Architected a full-stack TypeScript monorepo (pnpm workspaces) for a 700°C+ industrial aluminum furnace simulator, sharing strict Zod schemas and end-to-end types across a React/Zustand 5-stage operational state machine and a Fastify/Prisma/PostgreSQL backend.
- Engineered a low-latency WebSocket telemetry pipeline and deterministic scoring engine evaluating PPE compliance, charge inspection, thermal control, skimming, and emergency shutdown in real time, earning a 10/10 capstone defense score.

Memory Lens | Cognitive Computing Platform & 3D Neural Interface

GitHub

TypeScript, React 19, Three.js, React Three Fiber, WebGL/GLSL, Express, Gemini API, Vitest

- Engineered a modular cognitive memory architecture featuring custom Consolidation, Resonance, Forgetting (Ebbinghaus curve), and Blind-Spot Detection engines, cutting LLM token storage overhead by up to 80%.
- Built an Episodic Memory and Identity Evolution tracker alongside a 3D bioluminescent neural organism ("Chief") in React Three Fiber with custom GLSL shaders reactive to real-time cognitive state transitions.
- Conducted formal academic research on semantic document organization under Prof. Dr. Fabrício Londero (UniFacens), targeting submission to the ACDSA international conference.

PROFESSIONAL EXPERIENCE

CBI (Companhia Brasileira de Infraestrutura) 07/2026 - Present

Software Development Consultant | São Paulo, Brazil (Remote) (10/2026 - Present)

- Promoted to Software Development Consultant leading enterprise software engineering, internal platform architecture, and autonomous AI systems for mission-critical operations.
- Architected and deployed proprietary platforms on GCP (Cloud Run, Cloud SQL, Secret Manager, Cloud Build), including the **CBI Operations Center** (enterprise ITSM, Fleet Telemetry Radar, and Incident Correlation cutting projected TCO by 80% to 92%+) and **Onboarding OS** (multi-brand employee lifecycle platform with PostgreSQL Row-Level Security and AES-256-GCM encryption, designed for 6 public concessions and active in live pilot across CBI and municipal subsidiaries).
- Conceived and deployed **"Isa"**, a confined AI deflection agent (Google Gemini, MCP Servers, hybrid RAG, Zod JSON Schema validation) with predictive pre-ticket deflection, automated SLA routing, and automated RLS security guard tests, projecting R$ 32K to R$ 114K/year in operational savings.
- Direct technical engagements and architecture alignment with enterprise cloud vendors (Google Cloud and Microsoft) while safeguarding sensitive citizen data under strict LGPD compliance.
- Conduct corporate technical enablement workshops and establish AI governance frameworks, training engineering and operational teams on secure LLM workflows.

Full-Stack Developer | São Paulo, Brazil (Remote) (07/2026 - 10/2026)

- Engineered full-stack web applications and internal tools using TypeScript, React 19, Next.js, Node.js, and PostgreSQL.
- Implemented secure API integrations, webhooks, and automated security monitoring pipelines connecting enterprise platforms with Microsoft Graph API, Entra ID, and Intune endpoint compliance telemetry.
- Developed custom automation scripts in Python and TypeScript to optimize cross-departmental operations and incident deflection, resulting in promotion to Software Development Consultant.

Farmamed Drogarias 10/2025 - 03/2026

IT Infrastructure Analyst | Sorocaba, Brazil

- Managed corporate IT infrastructure, identity governance, access provisioning, and endpoint compliance across Microsoft Entra ID, Microsoft Intune, and Active Directory across multi-site retail branches.
- Developed Python automation scripts and data consolidation pipelines for log analytics, network diagnostics, and operational synchronization.
- Monitored mission-critical services and resolved high-priority technical incidents under strict operational SLAs.

Wobben Windpower (ENERCON Group) 11/2023 - 04/2025

Software & IT Systems Intern | Sorocaba, Brazil

- Participated in the global Microsoft Copilot Flight Crew program organized by Global IT, evaluating generative AI enterprise features and real-world adoption workflows.
- Maintained and executed PowerShell deployment scripts to streamline standardized OS provisioning, ensuring full endpoint compliance with ENERCON global headquarters standards.
- Supported enterprise administration across SAP ERP, Active Directory, and Microsoft 365, handling internal corporate technical support tickets via ServiceNow.

EDUCATION

Higher Technology Degree in Systems Analysis and Development (TADS) 2026

Centro Universitário SENAC, Brazil | Official Degree Average: 8.20 / 10.0 (US GPA: 3.38 / 4.0)

TECHNOLOGY RESIDENCY & APPLIED RESEARCH

ICT Resident Researcher (Digital Twins, XR & 5G) 10/2026 - 04/2027

UniFacens | Softex | Ministry of Science, Technology and Innovation (MCTI) | ICT Residency 51 (PPI)

- Ranked **8th nationwide out of 300+ candidates** in Phase 1: an intensive 360-hour specialization across AI, IoT, Extended Reality (XR), and 5G with a **4.00 / 4.0 GPA** (9.91 / 10.0 average, 10/10 in 10 modules), securing one of only 20 federal research fellowships awarded.
- Selected for Phase 2: a 6-month Technological Residency conducting applied research and enterprise PoCs across **Digital Shadow, Computer Vision, and Systems Integration**.
- Served as technical lead across 3 out of 4 collaborative capstone projects and lead presenter for 2 technical board defenses, including the FundiVR industrial digital twin simulator (approved with 10/10).

CERTIFICATIONS & LANGUAGES

Certifications: Google Cloud Generative AI Leader | Azure Well-Architected Framework & Cloud Architecture (Fast Lane / Microsoft) | Certified AppSec Practitioner (CAP) | Cisco CCNA 1 & Network Security

Languages: English (C1+ / Fluent, 2-Year Full Immersion at Happy & Co., Speaking: 4.5/5, Listening: 4.4/5) | Spanish (Intermediate / Technical) | Portuguese (Native)

Teaching & Mentoring: Former English Teacher & Mentor at Clube Poliglota Sorocaba, conducting cross-cultural speaking sessions for local and international attendees (including Latin America and Southeast Asia).